

KARTIK NEMANI PhD.Mechanical engineering (**minor: materials science**)

snemani@purdue.edu

Website: kartiknemani.github.io**LinkedIn:** Kartik Nemani**Education**

PhD in Mechanical Engineering, **Purdue University**

July 2019 – Sept 2024

Thesis: *Two-dimensional transition metal carbides as precursor materials for applications in ultra-high temperature ceramics*

Thesis Committee: Prof. Babak Anasori, Prof. Yury Gogotsi, Prof. Rodney Trice, Prof. Jing Zhang

M.S. in Mechanical Engineering, **University of Toledo**

Aug 2016 – Dec 2018

Thesis: *Engineered Thin Films for Applications in Organic Electronic and Photonic Devices*

Thesis Committee: Prof. Hossein Sojoudi, Prof. Yakov Laptisky, Prof. Sarit Baduri

Bachelor of Technology (B.Tech) in Mechanical Engineering, **Jawaharlal Nehru Technological University, Kakinada**

Oct 2010 – Aug 2014

Professional Experience

Post Doctoral Fellow

Feb 2025- present

University of Alabama, Birmingham

Post Doctoral Researcher

Sept 2024-Jan 2025

Purdue University, West Lafayette

Project Engineer

Sept 2015-Sept 2016

Unitech, India

Project Associate

Jan 2015-Sept 2015

Indian Institute of Technology, India

Awards/Recognitions

- **Web of Talents STEM Award-2nd Place (amongst 500+ finalists worldwide)** 2025
- Best poster award-American Ceramic Society (Engineering Ceramics Division) 2025
- Finalist (top 30) The Washington Research Fellowship (declined) 2024
- **American ceramic society graduate excellence in materials science (GEMS)-diamond award** 2024
- **University elite 50 award (Among 17 schools, 150 nominees, only award from school of engineering)** 2024
- Electron Microscopy awards: multiple platforms (JEOL, Nanoart, ACerS) 2020, '21, '22, '23, '24
- **Outstanding masters' thesis award, The University of Toledo** 2018
- National graduate aptitude test in engineering (GATE) qualified in~1,25,000 candidates (INDIA) 2015
- **Rank 2nd in the department of mechanical engineering- (graduating class of 72)** 2014

- Indian Institute of Technology-Kharagpur undergraduate engineering internship (Dr. M. Ramgopal) 2013
- Best paper award: Jawaharlal Nehru Technological University 2013
- Innovative rocket design award (Andhra University) 2012
- **Academic excellence: Ranked 1st in metallurgy and materials science** 2011

Travel Grants

Andhra University, National Science Day travel grant (2024), ACerS ECD grant (2022,2023), MRS travel award (2022), Purdue graduate school travel grant (2022), ECerS school travel grant (2022-declined), ACerS EMD travel grant (2021), ACerS-President’s council of student advisors travel grant (2021,2022, 2023).

Funding/Grants

Research

- | | | |
|---|-----------|------|
| - Web of Talents award | 1000 USD | 2025 |
| - ACerS GEMS award | 100 USD | 2024 |
| - ACerS-engineering ceramics division best poster | 200 USD | 2024 |
| - Purdue graduate fellowship | 60000 USD | 2019 |
| - University of Toledo graduate scholarship | 30000 USD | 2016 |

Teaching

- | | | |
|---|----------|-----------|
| - Teaching Assistantship- Purdue University | 5000 USD | 2022 |
| - Teaching Assistantship-University of Toledo | 7000 USD | 2017,2018 |

Provisional Patents

- | | |
|---|---------------------------|
| MXenes-metal and ceramic assemblies and composites
B Anasori, BC Wyatt, S.K. Nemani | US Patent App. 17/960,968 |
| High entropy MXenes and methods of making thereof
B Anasori, W Hong, S.K. Nemani | US Patent App. 17/497,026 |

List of **Publications** [\[Google Scholar\]](#) Citations: **2950** | h index: **17** | i-10 index: **21**

Keywords: Nanomaterials, 2D ceramics, high temperature materials, metal & ceramic composites, ultra-high temperature ceramics, polymer derived ceramics.

Only peer-reviewed articles

***co-first author** | *Italics: Journal name*
Studies on high temperature materials **highlighted**

Yet to be published

- Understanding phase and microstructure evolution of Ti₃C₂T_x MXene-polymer derived silicon carbide**
(Under Review, 2025).

2. **Room temperature stability and high temperature phase behavior of cations decorated $\text{Ti}_3\text{C}_2\text{T}_x$ MXene** (Submitted, 2025).
3. **A self-assembly approach for making 2D transition metals carbides-ceramics green bodies** (in preparation, 2026).

Published

2025

4. Cation exchange and systematic evaluation of $\text{M}_4\text{C}_3\text{T}_x$ MXenes for hydrogen evolution reaction electrocatalysis
A Thakur, Nithin Chandran BS, B. C. Wyatt, Priyanka Gurdev Singh, Annabelle Bedford, **S.K. Nemani**, Majid Beidaghi, Babak Anasori *Chemistry of Materials* 2025. (in press)
Doi: [10.1021/acs.chemmater.5c01706](https://doi.org/10.1021/acs.chemmater.5c01706)
5. **Order to disorder transition due to entropy in layered and 2D carbides**
B. C. Wyatt, Y. Yang, P. P. Michałowski, T. Parker, Y. Morency, F. Urban, G. Kadagishvili, M. Tanwar, S. P. Muhoza, **S.K. Nemani**, A. Bedford, H. Fang, Z. D. Hood, J. Jang, K. Kamath, B. G. Wright, R. Disko, A. Thakur, S. Han, N. Ghosh, X. Xu, Z. Fakhraai, Y. Gogotsi, A. Vojvodic, D.-e. Jiang, B. Anasori *Science* (2025), 389.6764. Doi: [10.1126/science.adv4415](https://doi.org/10.1126/science.adv4415).
6. **MXene-derived TiB_2 formation in B_4C at high temperatures**
Chandran BS, N., Thakur, A., **S.K. Nemani**, Moharana, N., Scott, C. P. H., Kumar, K. C., ... & Anasori, B. *Advanced Composites and Hybrid Materials*, (2025). 8(4), 1-17. Doi: [10.1007/s42114-025-01387-x](https://doi.org/10.1007/s42114-025-01387-x).
7. **$\text{Ti}_3\text{C}_2\text{T}_x$ MXene-zirconium diboride based ultra-high temperature ceramics**
S.K. Nemani, N. Gilli, S. Goldy, A. Kumar, Y. Im, A. J. Vorhees, B.C. Wyatt, N. Chandran BS, N. Chawla, G.J.Tucker, L.Silvestroni, B.Anasori, *Advanced Science*, (2025). Doi: [10.1002/advs.202500487](https://doi.org/10.1002/advs.202500487).

2024

8. A perspective on MXene-enhanced biofiltration-membrane water reuse treatment systems: A review and experimental validation
Firouzjaei, M. D., Clayton, J., Jafarian, H., Shamsabadi, A. A., Thakur, A., Todd, R., **S.K. Nemani**... & Terry, L, *Desalination*, (2025), 593, 118198.
Doi: [10.1016/j.desal.2024.118198](https://doi.org/10.1016/j.desal.2024.118198).

Book Chapter

9. Guidelines on fluorine-based synthesis of MXenes
Bedford, A., Wyatt, B.C., Highland, W.J., **S.K. Nemani**, Pulley, K.K., Jordan, J., & Anasori, B., (2024). *Transition Metal Carbides and Nitrides (MXenes) Handbook: Synthesis, Processing, Properties and Applications*, pp.34-57. John Wiley & Sons, Inc. Hoboken, NJ, USA. Doi: [10.1002/9781119869528.ch3](https://doi.org/10.1002/9781119869528.ch3).
10. **A generalized and scalable self-assembly method of 2D carbides to metal powders**
Patenaude, J., Wyatt, B. C., **S.K. Nemani**, & Anasori, B, *MRS Advances*, (2024), 9(9), 551-556. Doi: [10.1557/s43580-024-00854-9](https://doi.org/10.1557/s43580-024-00854-9).

11. Alkali cation stabilization of defects in 2D MXenes at ambient and elevated temperatures

Wyatt, B., Boebinger, M., Hood, Z., Adhikari, S., Michalowski, P., **S.K. Nemani**, Muraleedharan, M.G., Bedford, A., Highland, W., Kent, P., & others, *Nature Communications*, (2024), 15(1), 6353. [Doi: 10.1038/s41467-024-50713-2](https://doi.org/10.1038/s41467-024-50713-2).

2023

12. Metal-organic framework/Ti₃C₂T_x MXene-derived functional nanostructures for high-performance supercapacitors

Ahmad, R., Iqbal, N., Noor, T., **S.K. Nemani**, Zhu, L., & Anasori, B., *ACS Applied Nano Materials*, (2023), 7(1), pp.253-266. [Doi: 10.1021/acsanm.3c04389](https://doi.org/10.1021/acsanm.3c04389).

13. *Ultra-high temperature ceramics for extreme environments

S.K. Nemani, Brian C. Wyatt, Gregory E. Hilmas, Elizabeth J. Opila, Babak Anasori, *Nature Reviews Materials*, (2023), 9, 773–789. [Doi:10.1038/s41578-023-00619-0](https://doi.org/10.1038/s41578-023-00619-0).

14. Life-Cycle assessment of Ti₃C₂T_x MXene synthesis

Mostafa Dadashi Firouzjaei, **S.K. Nemani**, Mohtada Sadrzadeh, Evan K. Wujcik, Mark Elliott, Babak Anasori, (2023), *Advanced Materials*, 35 (31), 2300422. [Doi: 10.1002/adma.202370217](https://doi.org/10.1002/adma.202370217).

15. Functional two-dimensional high-entropy materials

S.K. Nemani, Torkamanzadeh M., Brian Wyatt, Volker Presser, Babak Anasori, *Nature Communications Materials*, (2023), 4 (1), 16. [Doi: 10.1038/s43246-023-00341-y](https://doi.org/10.1038/s43246-023-00341-y).

2022

16. Elucidating the chemical order and disorder in high-entropy MXenes: A high-throughput survey of the atomic configurations in TiVNbMoC₃ and TiVCrMoC₃

Z Leong, H Jin, ZM Wong, **S.K. Nemani**, B Anasori, TL Tan, (2022,) *Chemistry of Materials*, 34(20), 9062-9071. [Doi: 10.1021/acs.chemmater.2c01673](https://doi.org/10.1021/acs.chemmater.2c01673).

17. Ti₃C₂T_x MXene polymer composites for anticorrosion: an overview and perspective

I. Amin, H. Brekel, **S.K. Nemani**, E. Batyrev, A. de Vooys, H. van der Weijde, ..., *ACS Applied Materials & Interfaces*, (2022), 14 (38), 43749-43758. [Doi: 10.1021/acsami.2c11953](https://doi.org/10.1021/acsami.2c11953).

2021

18. High entropy 2D carbide MXenes: TiVNbMoC₃ and TiVCrMoC₃

S.K. Nemani, B. Zhang, B. Wyatt, Z. Hood, S. Manna, R. Khaledialidusti, W. Hong, M.G. Sternberg, S. Sankaranarayanan, B. Anasori, *ACS Nano*, (2021), 5 (8), 12815-12825. [Doi: 10.1021/acs.nano.1c02775](https://doi.org/10.1021/acs.nano.1c02775).

19. Ti₃C₂T_x MXene-polymer nanocomposites and their applications

Riazi, H., **S.K. Nemani**, Grady, M.C., & Anasori, B., *Journal of Materials Chemistry A*, , (2021), 9(13), pp.8051-8098. [Doi: 10.1039/D0TA08023C](https://doi.org/10.1039/D0TA08023C).

20. *2D transition metal carbides (MXenes) in metal and ceramic matrix composites

S.K. Nemani, Wyatt, Brian C., Babak Anasori, *Nano Convergence*, , (2021), 8(1), 16. [Doi: 10.1186/s40580-021-00266-7](https://doi.org/10.1186/s40580-021-00266-7).

21. ***High temperature stability and phase transformations of titanium carbide ($\text{Ti}_3\text{C}_2\text{T}_x$) MXene**

S.K. Nemani, Brian Wyatt, Krishay Desai, Harpreet Kaur, Bowen Zhang, and Babak Anasori, *Journal of Physics: Condensed matter*, (2021), 33 (22), 224002.

Doi: [10.1088/1361-648X/abe793](https://doi.org/10.1088/1361-648X/abe793).

2020

22. *Double transition metal MXenes: atomistic design of two-dimensional carbides and nitrides

S.K. Nemani, Weichen Hong, Brian Wyatt, and Babak Anasori, (2020), *MRS Bulletin*.

Doi: [10.1557/mrs.2020.251](https://doi.org/10.1557/mrs.2020.251).

23. Rational design of two-dimensional transition metal carbide/nitride (MXene) hybrids and nanocomposites for catalytic energy storage and conversion

Lim, K. R. G., Handoko, A. D., **S.K. Nemani**, Wyatt, B., Jiang, H. Y., Tang, J., ... & Seh, Z. W. (2020). *ACS nano*, 14(9), 10834-10864. Doi: [10.1021/acsnano.0c05482](https://doi.org/10.1021/acsnano.0c05482).

24. Delayed frost growth on nanoporous microstructured surfaces utilizing jumping and sweeping condensates

Mohammadian, B., Annavarapu, R. K., Raiyan, A., **S.K. Nemani**, Kim, S., Wang, M., & Sojoudi, H. (2020). *Langmuir*, 36(24), 6635-6650. Doi: [10.1021/acs.langmuir.0c00413](https://doi.org/10.1021/acs.langmuir.0c00413).

2019

25. Micro-/nanoscale approach for studying scale formation and developing scale-resistant surfaces

H Sojoudi, **S.K. Nemani**, K. M. Mullin, M.G. Wilson, H. Aladwani, H. Lababidi, ... (2019), *ACS applied materials & interfaces*, 11 (7), 7330-7337. Doi: [10.1021/acsami.8b18523](https://doi.org/10.1021/acsami.8b18523).

2018

26. Barrier performance of CVD graphene films using a facile P3HT thin film optical transmission test

S.K. Nemani, H. Sojoudi, (2018), *Journal of Nanomaterials*.

Doi: [10.1155/2018/9681432](https://doi.org/10.1155/2018/9681432).

27. Stretchable and hydrophobic electrochromic devices using wrinkled graphene and PEDOT:PSS

S.K. Nemani, D. Chen, M H Mohamad, H Sojoudi, (2018), *Journal of Nanomaterials*.

Doi: [10.1155/2018/3230293](https://doi.org/10.1155/2018/3230293).

28. Surface modification of polymers: methods and applications

S.K. Nemani, Annavarapu, R. K., Mohammadian, B., Raiyan, A., Heil, J., Haque, M. A., ... & Sojoudi, H. (2018), *Advanced Materials Interfaces*, 5(24), 1801247.

Doi: [10.1002/admi.201801247](https://doi.org/10.1002/admi.201801247).

Selected Conferences and Invited Talks

- Lifecycle assessment of 2D transition metal carbides (MXenes), and their use as functional additives in ceramics and electrocatalytic applications (invited)

(Global Young Investigator Forum)

American Ceramic Society-ICACC 2025

- 2D transition metal carbides (MXenes) and their applications (invited)

National Science Day Lecture Series
Andhra University, Feb 2024

- Next-gen nanomaterials for futuristic technologies (invited)

Jawaharlal Nehru Technological University
Gayatri College of Engineering, 2024

- MXene derived carbides as precursors for UHTCs
(**GEMS-Diamond award**)

American Ceramic Society-MS&T 2024

- Discovery of multiple principal element (high entropy) MXenes

Materials Research Society
Spring 2022

- High-temperature stability and phase transformations of 2D titanium carbide MXene

American Ceramic Society-MS&T 2021

Over 10 conference presentations (poster and oral) in national and international forums including American Ceramic Society, Materials Research Society, Vacuum Association, American Chemical Society, Chinese Academy of Sciences, European Ceramic Society, American Physical Society, American Electrochemical Society, and Royal Chemical Society.

Grant Writing Experience

I have assisted in drafting the following grant proposals and white papers: * indicates funded

1. Advanced Characterization of Post-Irradiation Effects in Polymer-Derived SiC and C-ZrC-SiC Composites
PI: Kathy Lu, Co-investigator: Kartik Nemani
Agency: NSUF-Super RTE (Nuclear Science User Facility-Super Rapid Turnaround Experiment) 2025*
2. Two-dimensional Transition Metal Carbides-Silicon Carbide Interfaces for Tunable Electronic & Thermal Properties
PIs: Babak Anasori, Laura Silvestroni
Funding Agency: European Industry collaborative 2025
3. Development of Lightweight Multi-band Electromagnetic Wave Absorbing Materials Technology for Future Mobility
PI: Korea Electronics Technology Institute, Babak Anasori 2024*
Funding Agency: Ministry of Trade, Industry and Energy, South Korea
4. 2D Transition Metal Carbide MXene as an Additive in Silicon Carbide Ceramics 2024*
PI: Babak Anasori
Funding Agency: AFRL
5. In-situ annealing and XRD of MXene-reinforced ceramics and MAX phases 2022*
PI: Babak Anasori
Funding Agency: DURIP-Office of Naval Research (ONR), USA
6. 2D MXenes Ultra-high Temperature Materials for Hypersonic Systems 2020*
PI: Babak Anasori
Funding Agency: Office of Naval Research (ONR), USA

7. Out-of-Autoclave Composite Curing and Repair Utilizing Graphene Nanostructured Heaters 2018
 PIs: Nagaraja Iyyer, Hossein Sojoudi, Mehdi Naderi
 Funding Agency: Department of Defense (DOD), USA

Leadership, Mentoring, and Outreach

- Chair**-professional development committee-The president's council of student advisors (PCSA)
 American Ceramic Society (ACerS) 2021-2023
- Member**-(Young professionals board) education and professional development council-ACerS 2021- NA
- Co-Founder**- Nano Science and Engineering Association (NSEA) 2020
- Multi-Level Mentor**
- Multi-Disciplinary Undergraduate Research Institute (MURI) Mentorship 2020-2024
 - Engineering Innovation & Leadership (NEIL) scholar mentor 2023
 - Undergraduate capstone projects- Purdue University 2022,2023
 - National Science Foundation-Research Experience for Undergraduates (NSF-REU) program 2022-2024
 - Undergraduate mentoring- University of Toledo 2017-2018

List of Student mentees

University of Toledo		Purdue University	
- Jamie Heil (BS/MS)*	2017-2018	- Indranil Roy (BS)	2020
- Marwan Mohamed (BS)*	2017	- Krishay Desai (BS)*	2021
- Corey Williams (BS)	2017	- Yooran Im (BS)* ^{1,3}	2021-2022
- Aaron Grossweiler (BS)	2017	- Austin Vorhees (BS/MS)* ²	2022-2024
		- Isaac Fritz (BS)	2023-2024
		- Krutarth Kamath (PhD)*	2023-2024
University of Alabama			
- Shilpa C. (MS/PhD)	2025		

* Co-Authors on peer-reviewed publications

1. recipient of MURI scholarship (Multidisciplinary Undergraduate Research Institute)
2. recipient of NEIL scholarship (Navy Engineering Innovation & Leadership)
3. recipient of UROP scholarship (Undergraduate Research Opportunity Program)

Science Advocacy & Outreach

- *Science Magazine* NexGen Voices- News from the Post Pandemic World Vol 369 ISSUE 6499 2020
- *ACerS Bulletin*- featured article on "Hope and Science" Vol 101 ISSUE 9 2022
- 2D MXenes Ink: Synthesis, Properties, and Their Precursors (sigmaaldrich.com) 2022
- MRS Volunteer: contact your government representative initiative 2022
- Hypersonic flight: Redefining the aerospace frontier - The American Ceramic Society 2025

Teaching

- **Thermodynamics II (MIME 341)** [3 credit Hours]
 Spring 2018, University of Toledo

- **Heat Transfer (MIME 3440)** [3 credit Hours]
Fall 2017, Summer 2018, University of Toledo
- **Introduction to Nanotechnology (ME 423)** [3 credit Hours]
Fall 2023, Purdue University

Labs and Workshops

- **Control Systems Analysis and Design (ME 482)** [3 credit Hours]
Fall 2022, Purdue University
- **NanoSystems Principles (ME 523)** Scanning Electron Microscopy/ X-ray Diffraction [3 credit Hours]
Spring 2024, Purdue University

Service

Guest Editor

Ceramics (Nanoceramics and Two-Dimensional Ceramic Materials) 2025

Reviewer

- | | |
|---|---|
| - Nature Communications | - Frontiers in Materials |
| - Carbon (Elsevier) | - SN Applied Sciences (Springer Nature) |
| - Journal of the European Ceramic Society (ECerS) | - Surfaces and Interfaces (Elsevier) |
| - Graphene and 2D Materials (Springer Nature) | - Advanced Materials (Wiley) |
| - Nature Scientific Reports (Springer) | - Optical Materials (Elsevier) |
| - Applied Materials & Interfaces (ACS) | - Heliyon (Elsevier) |
| - Reviews in Chemical Engineering (DeGruyter) | - Surfaces and Interfaces (Elsevier) |
| - Nano-Micro Letters (Springer) | - International Journal of Applied Ceramic Technology (ACerS-Wiley) |
| - Journal of Electrochemical Energy Conversion and Storage (ASME) | - Next Materials (Elsevier) |
| | - Physica Scripta (IOP) |

Symposium Organizer

- IGNITE MSE Professional Development Symposium- MS&T 2023, 2024
- IGNITE MSE Workshop- ICACC 2024

Conference Session Chair

- Young Investigators Forum (ICACC, 2025)
- MXenes Symposium (MRS Spring 2022, Fall 2022)
- Advanced Materials for harsh environments (MS&T Fall 2021)

Media Coverage |These are hyperlinks|

1. First sample of MXene-enhanced UHTCs confirms potential of this advanced materials design pathway - The American Ceramic Society 2025
2. Hypersonic flight: Redefining the aerospace frontier - The American Ceramic Society 2025
3. AFOSR on X: "Autumn's Final Bloom: The Last Rose Standing" AFOSR, Art of Science Showcase 2023

4. New frontiers in space travel: A review of ultrahigh-temperature ceramic properties and processes - The American Ceramic Society 2023
5. University of Maine: Article by Purdue University, University of Alabama & University of Alberta (researchers published on MXene production environmental impact) 2023
6. Purdue University School of Engineering and Technology: Student Successes 2022
7. A new frontier for 2D materials—researchers create high-entropy MXenes - The American Ceramic Society 2021
8. Newly discovered high-entropy MXenes have high-tech applications (nanowerk.com) 2021
9. [New high-tech nanomaterials: Research Impact: Indiana University](#) 2021

Professional Memberships

- American Ceramic Society
- Materials Research Society
- Global Graduate Researchers Network
- Institution of Engineers (India)
- Materials Advantage